

Mail Server

Postfix + Dovecot · Linux System & Network Administration

1. Service Fundamentals

A Linux mail stack typically combines an MTA (Postfix) that sends and receives SMTP mail with a mail-access server (Dovecot) that lets clients retrieve mail via IMAP or POP3. Optional webmail provides a browser interface.

Component	Software	Role / Port
MTA	Postfix	Send/receive · SMTP 25, Submission 587
Access	Dovecot	IMAP 143/993 · POP3 110/995
Client	Thunderbird, Outlook	User agent

Item	Details
Postfix Config	/etc/postfix/main.cf · master.cf
Dovecot Config	/etc/dovecot/dovecot.conf · conf.d/
Mail Store	~/Maildir or /var/mail/
Logs	/var/log/mail.log · journalctl -u postfix

2. Architecture

Outbound: Client → Submission (587) → Postfix → remote MX.
Inbound: Remote MTA → SMTP 25 → Postfix → local delivery → Client retrieves via IMAP/POP3.

3. Installation

Update package list and install the mail stack:

Command:

```
sudo apt update
```

Command:

```
sudo apt install postfix dovecot-imapd dovecot-pop3d
```

During Postfix installation choose "Internet Site" and set the system mail name.

Enable and start both services:

Command:

```
sudo systemctl enable --now postfix dovecot
```

4. Configuration

4.1 Postfix (main.cf)

Command:

```
sudo nano /etc/postfix/main.cf
```

Key settings:

```
myhostname = mail.company.local
mydomain = company.local
myorigin = $mydomain
inet_interfaces = all
mydestination = $myhostname, localhost.$mydomain, $mydomain
home_mailbox = Maildir/
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
```

4.2 Dovecot

Set the mail location (usually in conf.d/10-mail.conf):

Command:

```
sudo nano /etc/dovecot/conf.d/10-mail.conf

mail_location = maildir:~/Maildir
```

Enable authentication (conf.d/10-auth.conf):

Command:

```
sudo nano /etc/dovecot/conf.d/10-auth.conf
```

```
disable_plaintext_auth = no
auth_mechanisms = plain login
```

In production set `disable_plaintext_auth = yes` and use TLS.

Apply the changes:

Command:

```
sudo systemctl restart postfix dovecot
```

5. Verification

Check both services:

Command:

```
sudo systemctl status postfix dovecot
```

Confirm the ports are listening:

Command:

```
sudo ss -lntp | grep -E ':25|:587|:143|:993'
```

Send a test message:

Command:

```
echo "Test body" | mail -s "Test" user@company.local
```

Watch the mail log:

Command:

```
sudo tail -f /var/log/mail.log
```

6. Configuration Reference

File / Port	Purpose
/etc/postfix/main.cf	Primary Postfix settings
/etc/dovecot/dovecot.conf	Dovecot main configuration
TCP 25	SMTP (inbound / MTA-to-MTA)
TCP 587	Submission (authenticated clients)
TCP 143 / 993	IMAP / IMAPS
TCP 110 / 995	POP3 / POP3S
~/Maildir/	Per-user mail storage (Maildir format)

7. Troubleshooting

Problem	Check	Solution
Mail not received	mail.log, MX, firewall 25	Fix DNS MX; open port 25; check mydestination
Relay access denied	smtpd_relay_restrictions	Authenticate or be in mynetworks
IMAP login fails	dovecot logs, Maildir	Create Maildir; fix password; enable auth
Service not listening	ss -lntp · systemctl	Restart postfix/dovecot; check inet_interfaces

8. Security

- Prefer submission on 587 with SASL + TLS; never run an open relay.
- Use IMAPS (993) or STARTTLS.
- Publish SPF, DKIM and DMARC records for public domains.
- Restrict mynetworks and keep Postfix/Dovecot updated.

9. Real-World Scenario

A small company wants @company.local mail. The administrator installs Postfix + Dovecot, creates system users, sets `home_mailbox = Maildir/`, publishes MX and A records, opens ports 25/587/993, and configures Thunderbird for IMAP + SMTP submission.

10. Essential Commands

Command	Purpose
sudo apt install postfix dovecot-imapd	Install the stack
sudo postconf -n	Show non-default Postfix settings

Command	Purpose
<code>sudo systemctl restart postfix dovecot</code>	Apply changes
<code>echo test mail -s subj user</code>	Send a test message
<code>sudo tail -f /var/log/mail.log</code>	Live mail log

11. Important Points (Interview / Viva)

1. MTA (Postfix) transfers mail; Dovecot provides IMAP/POP3 access.
2. Ports: 25 (SMTP), 587 (submission), 143/993 (IMAP), 110/995 (POP3).
3. Config: `/etc/postfix/main.cf` · `/etc/dovecot/`.
4. Maildir is preferred over mbox for concurrent access.
5. Open relay is dangerous — always require authentication or limit mynetworks.